

CHIJ St. Theresa's Convent
Additional Mathematics
2026 4E5N Preliminary Examination Paper 1

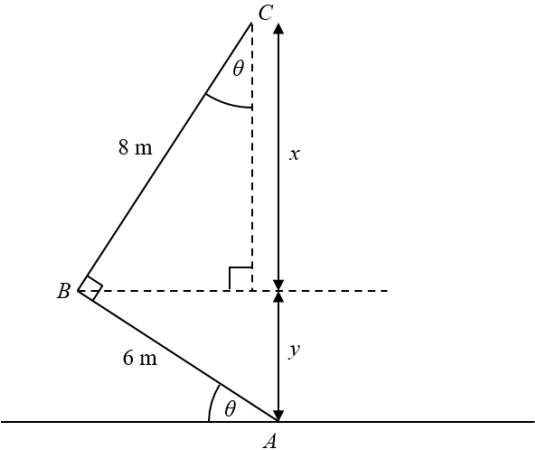
Marking Scheme

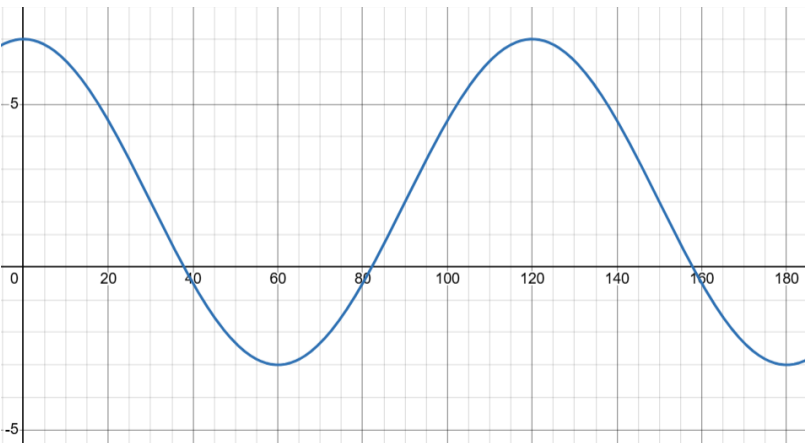
1		$\begin{aligned} \text{Base area} &= (3+2\sqrt{3})^2 \\ &= 21+12\sqrt{3} \\ \\ \text{Height} &= \frac{141+81\sqrt{3}}{21+12\sqrt{3}} \\ &= \frac{141+81\sqrt{3}}{21+12\sqrt{3}} \times \frac{21-12\sqrt{3}}{21-12\sqrt{3}} \\ &= \frac{45+9\sqrt{3}}{9} \\ &= 5+\sqrt{3} \\ \\ \text{Total surface area} &= 2(21+12\sqrt{3})+4(3+2\sqrt{3})(5+\sqrt{3}) \\ &= 126+76\sqrt{3} \end{aligned}$	M1 A1 M1 M1 A1 M1 A1	
2	(a)	$\begin{aligned} \frac{d}{dx} \sqrt{2x^2-3x+1} &= \frac{1}{2}(4x-3)(2x^2-3x+1)^{-\frac{1}{2}} \\ &= \frac{4x-3}{2\sqrt{2x^2-3x+1}} \end{aligned}$	B1, B1	B1 for numerator B1 for denominator
	(b)	$\begin{aligned} \int \frac{24x-18}{\sqrt{2x^2-3x+1}} dx &= 12 \int \frac{4x-3}{2\sqrt{2x^2-3x+1}} dx \\ &= 12\sqrt{2x^2-3x+1} + c \end{aligned}$	M1 A1	A0 if no +c

3		$\tan(A+B) = 6$ $\frac{\tan A + \tan B}{1 - \tan A \tan B} = 6$ $\frac{\tan A + 2}{1 - 2 \tan A} = 6$ $\tan A + 2 = 6 - 12 \tan A$ $13 \tan A = 3$ $\tan A = \frac{3}{13}$ $\text{Hypotenuse} = \sqrt{3^2 + 13^2}$ $= \sqrt{178}$ $\cos A = \frac{13}{\sqrt{178}}$ $\frac{13\sqrt{178}}{178}$	 M1 M1 A1 M1 A1	 M1 for substituting $\tan B = 2$ M1 for getting rid of denominator M1 for applying Pythagoras Theorem A0 if denominator not rationalised
4	(i)	Gradient of $AB = -1$ Equation of AB: $y + 2 = -(x - 1)$ $y = -x - 1$ $-x - 1 = x - 15$ $x = 7$ $\rightarrow y = -8$ $\therefore$ Coordinates of B is $(7, -8)$	 B1 M1 M1 A1	 M1 for finding equation of AB M1 for solving simultaneous equation using tangent at B and AB
	(ii)	$\text{Radius} = \frac{1}{2} \sqrt{(7-1)^2 + (-8+2)^2}$ $= 3\sqrt{2}$ $\text{Coordinates of centre} = \left(\frac{1+7}{2}, \frac{-2-8}{2} \right)$ $= (4, -5)$	 M1 A1 B1	 A0 if answer not simplified, i.e. $\sqrt{18}$

5	(i)	$(2-x)^2(1-px)^8$ $= (4-4x+x^2)(1-8px+28p^2x^2+\dots)$ $= 32px^2+x^2+112p^2x^2+\dots$ $112p^2+32p+1=81$ $7p^2+2p-5=0$ $(7p-5)(p+1)=0$ $p=\frac{5}{7} \quad \text{or} \quad p=-1$	B1 B1 M1 A1, A1	B1 for coefficient of x^2
	(ii)	Coefficient of $x^3 = \binom{8}{3}(-p)^3$ $= -56p^3$ When $p = \frac{5}{7}$, Coefficient = 56 When $p = -1$, Coefficient = $-\frac{1000}{49}$	M1 A1 A1	
6	(i)	Let $\angle BAE = x^\circ$ $\angle CBE = 2x^\circ$ (ext. $\angle$ of triangle) $\angle DCE = \angle CAE$ $= x^\circ$ (alt. seg theorem) $\angle CDE = (180-2x)^\circ$ (tangent @ ext. pt.) $(\angle \text{ sum of triangle})$ Since $\angle CBE + \angle CDE = 180^\circ$, B, C, D and E lie on a circle ($\angle$ in opp seg)	B1 B1 B1 B1, DB1	DB1 for reasoning
	(ii)	$\angle DCE = x^\circ$ $\angle EBD = \angle DCE$ $= x^\circ$ ($\angle$s in same seg) $\angle CBD = 2x - x$ $= x^\circ$ Since $\angle EBD = \angle CBD = x^\circ$, BD bisects $\angle CBE$ (shown)	B1 DB1	

7	(i)	$\frac{4}{3}\pi r^3 = 200 \times 60$ $r^3 = \frac{9000}{\pi}$ $r = 14.20248085$ $r = 14.2 \text{ cm (3sf)}$	M1 A1	
	(ii)	$V = \frac{4}{3}\pi r^3$ $\frac{dV}{dr} = 4\pi r^2$ $A = 4\pi r^2$ $\frac{dA}{dr} = 8\pi r$ $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dV} \times \frac{dV}{dt}$ $= 8\pi r \times \frac{1}{4\pi r^2} \times 200$ $= \frac{400}{r}$ When $r = 14.20248085$, $\frac{dA}{dt} = 28.1640936$ $= 28.2 \text{ cm}^2/\text{s (3sf)}$	B1 B1 M1 A1	
8	(i)	$2\pi r h + \pi r^2 = 320\pi$ $h = \frac{160}{r} - \frac{1}{2}r$ $V = \pi r^2 h$ $V = \pi r^2 \left(\frac{160}{r} - \frac{1}{2}r \right)$ $V = 160\pi r - \frac{1}{2}\pi r^3 \text{ (shown)}$	B1 M1 AG1	

	(b)	$kx^2 + 6x + k - 1 = -9$ $kx^2 + 6x + k + 8 = 0$ For no real roots, $D < 0$ $6^2 - 4(k)(k+8) < 0$ $k^2 + 8k - 9 > 0$ $(k+9)(k-1) > 0$ $k < -9$ or $k > 1$ (reject)	B1 M1 A1 M1 A1	M1 for discriminant M1 for factorising
10	(i)	 $L = x + y$ $= 8 \cos \theta + 6 \sin \theta$	M1 AG1	
	(ii)	$R = \sqrt{8^2 + 6^2}$ $= 10$ $\alpha = \tan^{-1}\left(\frac{6}{8}\right)$ $= 36.86989765^\circ$ $= 36.9^\circ$ (1dp) $L = 10 \cos(\theta - 36.9^\circ)$ Max $L = 10$ When $\theta - 36.9^\circ = 0$ $\therefore \theta = 36.9^\circ$ (1dp)	B1 M1 A1 A1 B1 DB1	
	(iii)	$10 \cos(\theta - 36.870^\circ) = 7$ $\cos(\theta - 36.870^\circ) = 0.7$ Basic $\angle, \alpha = \cos^{-1}(0.7)$ $= 45.572996^\circ$ $\therefore \theta = 82.4289365^\circ$ $= 82.4^\circ$ (1dp)	M1 A1	
11	(i)	Period $= 120^\circ$ or $\frac{2\pi}{3}$ rad	B1	

	(ii)	$\text{Amplitude} = \frac{7 - (-3)}{2}$ $= 5 \text{ units}$	B1	
	(iii)	$p = 5$ $q = 2$	B1 B1	
	(iv)	$5 \cos 3x + 2 = 0$ $\cos 3x = -0.4$ Basic $\angle, \alpha = \cos^{-1}(0.4)$ $= 66.42182152^\circ$ (2 nd & 3 rd quad) Smallest $x = 37.85939283^\circ$ $= 37.9^\circ$ (1dp)	M1 A1	
	(v)		P1 Q1 C1	P1 for correct number of cycles Q1 for correct max/min C1 for smooth curve and correct shape
12	(i)	$y = \sqrt{2x+7}$ $\frac{dy}{dx} = \frac{1}{\sqrt{2x+7}}$ When $x = 1$ $y = 3$ $\therefore B(1, 3)$ Gradient of $BC = -3$ Equation of BC : $y - 3 = -3(x - 1)$ $y = -3x + 6$ When $y = 0$ $x = 2$ Coordinates of $C = (2, 0)$	B1 B1 B1 B1	

	(ii)	When $y = 0$ $x = -\frac{7}{2}$ Coordinates of $A = (-3.5, 0)$ Area of shaded region $= \int_{-\frac{7}{2}}^1 \sqrt{2x+7} \, dx + \frac{1}{2}(1)(3)$ $= \left[\frac{(2x+7)^{\frac{3}{2}}}{\frac{3}{2}} \right]_{-\frac{7}{2}}^1 + \frac{3}{2}$ $= \left[\frac{(2+7)^{\frac{3}{2}}}{\frac{3}{2}} \right] - \left[\frac{(-7+7)^{\frac{3}{2}}}{\frac{3}{2}} \right] + \frac{3}{2}$ $= 10.5 \text{ units}^2$	B1 M1, M1 A1 M1 A1	M1 for finding area under curve M1 for finding area under straight line A1 for anti-derivative M1 for substitution of upper and lower limits
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